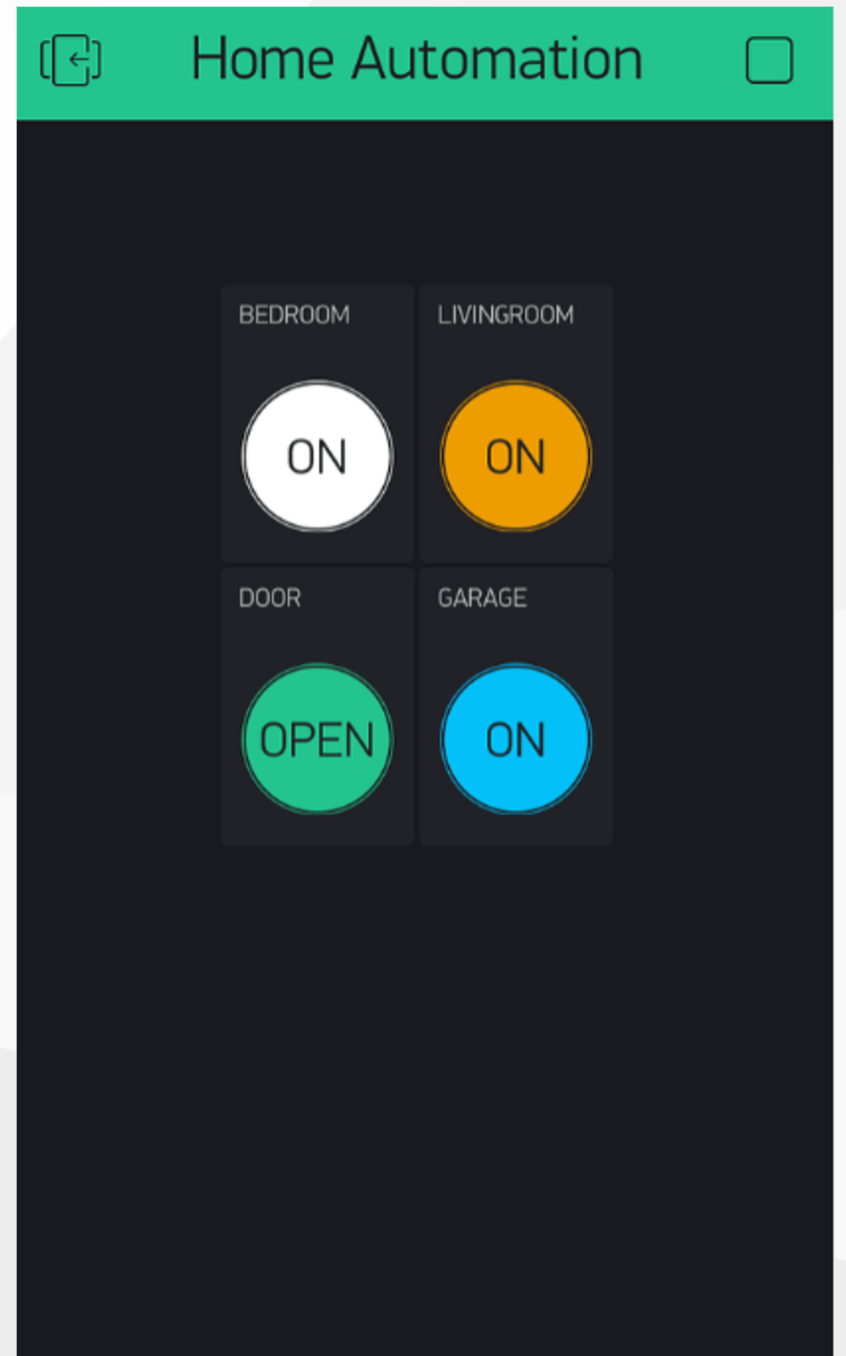


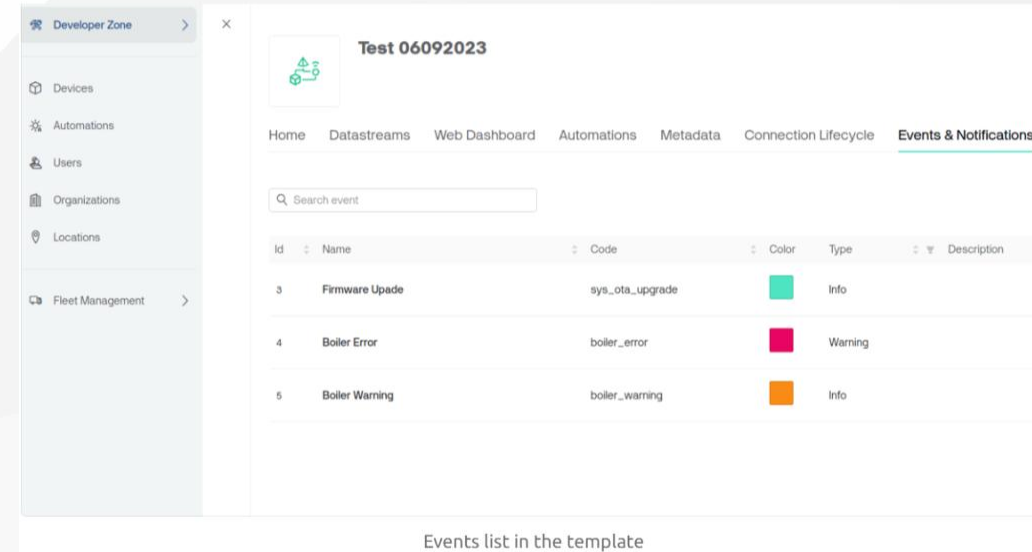
Blink Events and Automations





Blynk Events

- Events are used to track, log, and work with important events.
- Events are also used for notifications which can be sent over email, push notifications, or SMS.
- Events are pre-configured in Blynk.Console
- Can be triggered by:
 - Blynk Library/Firmware from the device
 - Using the Events HTTP API



Blynk Event Examples

- Create Warning Event when temperature reaches a certain threshold and send a notification
- Create Error Event status/malfunction of device
- Track the total working hours of the device. Notify tech support to provide maintenance
- Track online availability of device to comply with Service Level Agreement
- Unexpected Vibrations in Mechanical Device

(From Blynk Website)

Types of Events

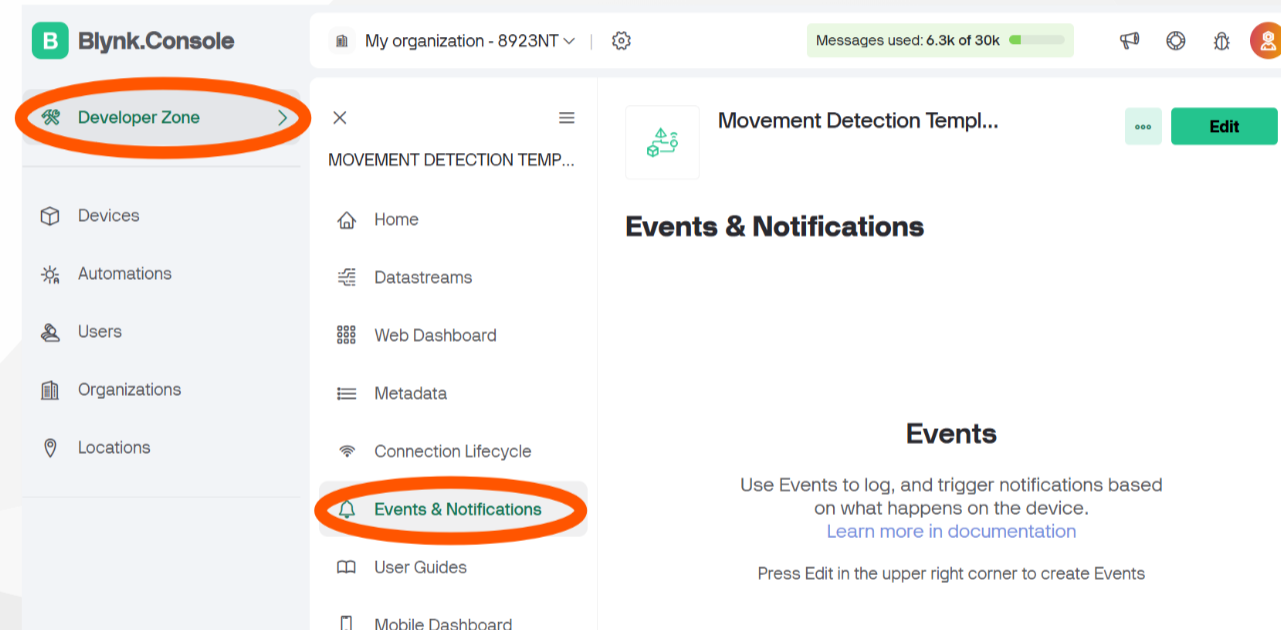
There are three types of Events in Blynk:

- **System Events:** Default Blynk platform events, like "OTA Update"
 - Unable to delete or edit system events. Example: Firmware Over-The-Air Update status
- **Custom Events:** Events you can create and configure for your needs
 - Events are specific to your application or device.
- **Content Events:** Informative events that are shown separately in the app

Event Definition in Blynk

- Open existing Template or create a new template in Blynk
- Go to Developer Zone -> My Templates -> Select a template -> Open the "Events & Notifications" tab.

Each event is assigned an `EVENT_CODE` , which is used in the firmware API or HTTPS API.



Configuring Events

- You need to explicitly configure Event settings :
 - Event frequency limits(every minute/hour etc)
 - Event visibility in Blynk App/console
 - Notification Configurations
 - Automation Access
 - Can it be used to initiate automations

test

General

Notifications

Settings

Expose to Automations

Limit

Every message will trigger the event

Event will be sent to user only once per

Show in Notifications & Events section:

☒ Device page (app and console)

☒ Home screen (app only)

Other

Apply tag when this event is recorded

Sending Events Example

Can Send Events using

1. Blynk Python Library
2. HTTP API

Create Events Using Python library

- Use the `blynk.logEvent(event_code)` command to trigger a new event:

```
if temperature > 35:  
    blynk.log_Event("high_temp")
```

- Executing this code will log the Event .
- Associated automations will occur (render on the timeline, send notifications, etc.)
- **WARNING: Don't log an event too many times to avoid hitting daily limits. 100 events per device per day.**

Setting Custom Event Description

- It is possible to change the description of the event when it's rendered on the timeline in Blynk.Console and in Blynk.Apps. For example, you can include the current data

```
if temperature > 35:  
    Blynk.logEvent("event_code", f"High TemperatureDetected! {temperature}°")
```

Create Events Using HTTP API

- Blynk allows access to Events via the Event HTTP API
 - Use the Query String of the URL to specify event code, auth token and optional description
- Log an event via GET request:

```
`/external/api/logEvent?token={AuthToken}&code={event_code}
```

- To add a custom description to the event, use this GET request

```
`/external/api/logEvent?token={AuthToken}&code={event_code}&description={event_description}
```

```
https://blynk.cloud/external/api/logEvent?token=GadsdadsavsV0YD3e14y00pneL1&code=firmware_update&description=test
```

Blynk Event using Python Library

```
Blynk.log_Event("event_code", "optional message");
```

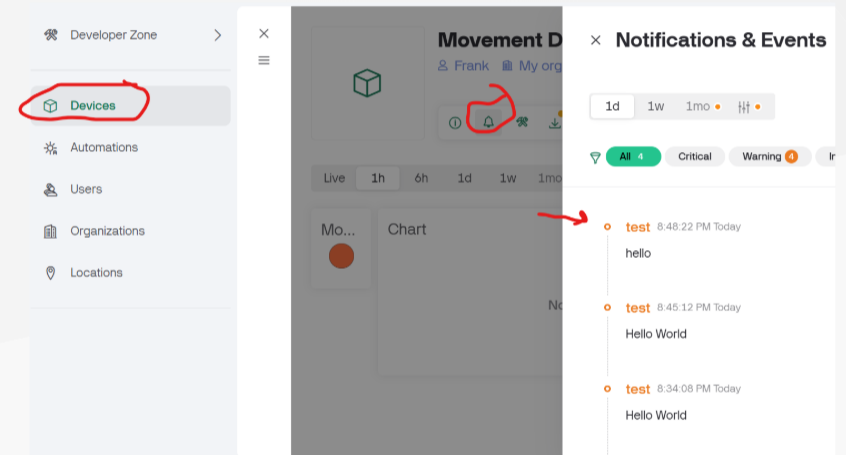
```
if some_condition:  
    blynk.log_Event("hello");  
}
```

Optionally, you can send a custom description of the event. This description will be rendered on Device Timeline.

```
if (some_condition){  
    Blynk.log_event("hello", "Hello World,") ;  
}
```

Event Log

- Blynk Console, click on the Bell icon to open the Notifications & Events console.



Event Notifications

You can also configure Push Notifications/email/sms to occur on an event:

- Open Template, go to Events, and enable via Notifications tab

button_pressed

General **Notifications** Settings

☒ Enable notifications

Default recipients

E-MAIL TO

Select contact

PUSH NOTIFICATIONS TO

Device Owner

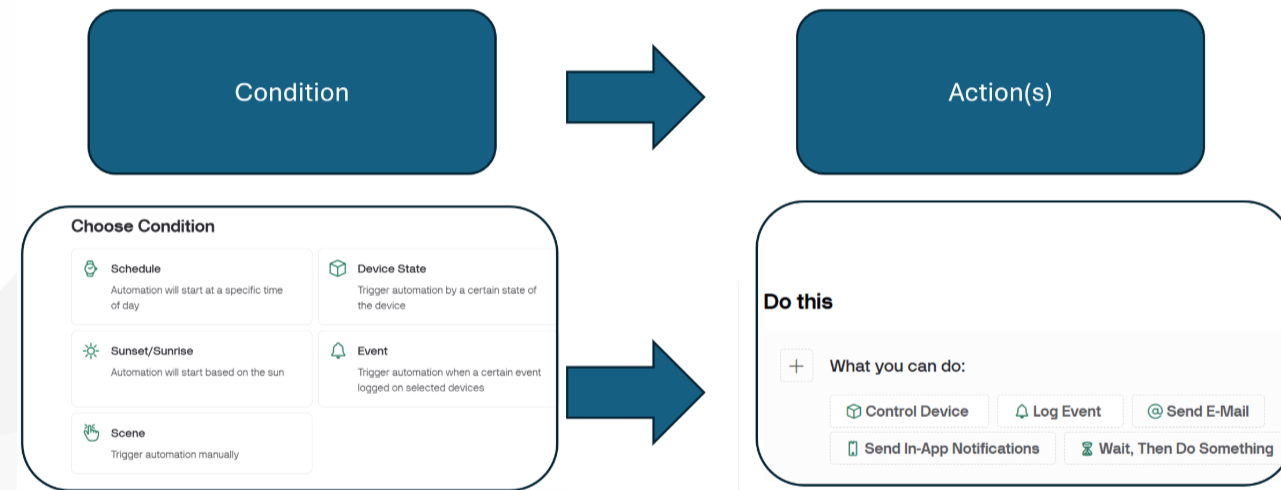
Events Considerations

- You can send only 100 events per device per day
- When the limit is reached you'll see the notification on the UI in the Device Timeline section
- The maximum description length for the event is 300 characters

Blynk Automations

Automations

- Enables the creation of scenarios where devices perform specific actions based on defined conditions, enhancing the functionality and responsiveness of IoT solutions.
- Automations configured through both the Blynk Console and Blynk App



Condition Types

- **Schedule:** Triggers automation at specified dates and times, considering the relevant time zone.
- **Sunrise/Sunset:** Initiates automation relative to sunrise or sunset times on selected weekdays at a specific geographic location.
- **Device State:** Activates automation based on the value of a datastream from a device.
- **Scene:** Allows manual triggering of automation scenarios directly from the Blynk.App or Blynk.Console.
- **Event** Triggers Automation when a certain event occurs

Types of actions

One or more actions may be executed when a condition's requirement has been met.

Examples:

- **Control Device:** Sets a datastream value for a specific device.
- **Forward Device Data:** Updates a datastream value for one device based on data from another device.
- **Send Email:** Dispatches an email to designated users with optional details about the organization, template, device, and datastream value.
- **Send In-App Notification:** Sends a notification within the Blynk.App to specified users, including optional details.
- **Wait, Then Do Something:** Introduces a delay until a specific time or for a set duration before executing the next action.
- **Send SMS:** Sends a text message to specified users; this option is available only to BUSINESS subscribers.

Why use automations: Interoperability

- Automations are external to devices.
 - Can work across all devices
- One condition can be used to perform actions on multiple devices.
- The value of a Datastream(e.g. temperature) may be used by a condition
- Automation can be configured to set a Datastream to a new value.
- Widgets on the Blynk.Console and/or Blynk.App, you can visualise and change the values of Datastreams.
- You may also access Datastreams within the firmware of an IoT device, and by using a HTTP API.

Set Up Automations in Blynk

Prepare Datastreams for use in Automations

- Go to Blynk.Console -> Templates -> Template -> Datastreams and select the datastream you wish to use
- Make a choice for the Datastream to be associated with your Automation

Virtual Pin Datastream



View mode

You are in a "View" mode. Tap on "Edit" button in the top right corner to make changes.

General

Expose to Automations

AUTOMATION TYPE

Switch



Condition

Allow users to use this datastream as an Automation condition



Action

Allow users to change this datastream's state using Automations

Close

Prepare Event for use in Automations

- Go to Blynk.Console -
> Templates -> Template ->
Events and select the event you
wish to use
- Edit the Event as follows:

Expose to Automations

CONDITION

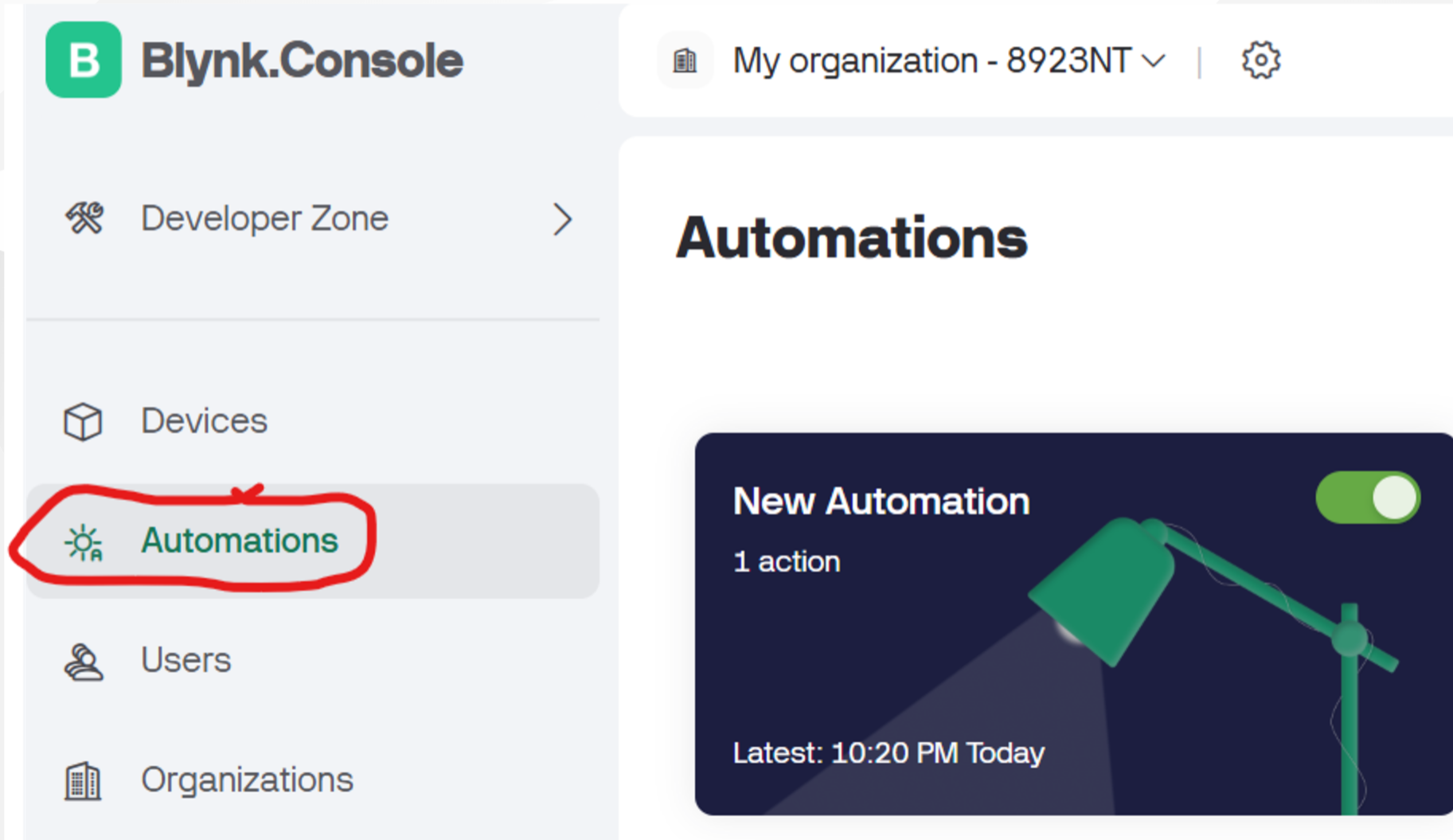


ACTION



Create an Automation

- From the [Blynk.Console](#), click on the Automations item on the main menu.



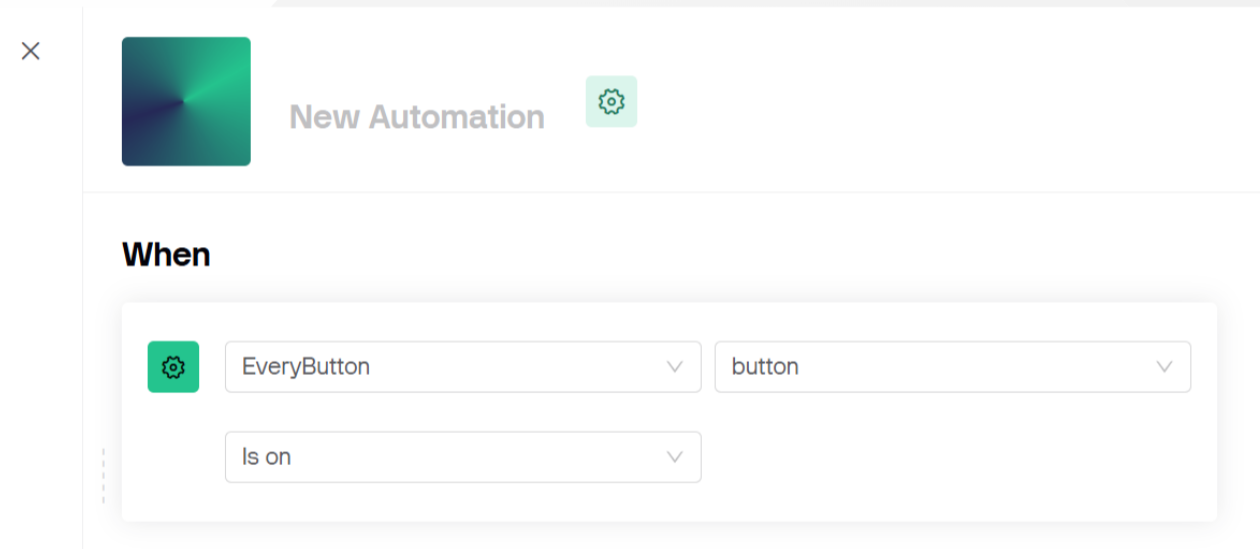
Placeholders

The 'Send Email', 'Send In-App Notifications', and 'Send SMS' actions support the use of placeholders. Placeholders are key/value pairs that refer to Blynk account metadata and [DataStream](#) values.

- {ORG_NAME} - name of the organization the device is assigned to.
- {PRODUCT_NAME} - name of the Template used.
- {DEVICE_NAME} - the IoT Device name as displayed under 'DEVICES'.
- {TRIGGER_VALUE} - if a Device State trigger was used, this will be the value of the trigger.

Configuring a Device State Condition

- The condition (When) option of 'Device State' triggers an action based on the value of a datastream.
- After selecting the 'Device State' option, you need to specify the device, then the datastream, and finally what type of datastream value change to monitor from the dropdown lists.



Automation Management

After the Automation is configured, it will appear in the 'Automations' list as a card, with the count of actions defined, and the last date/time that the Automation was executed. A switch on the card allows the Automation to be enabled/disabled.

Automations

Sort by:

Last Created



New Automation [1]

1 action

Not triggered yet



New Automation

1 action

Latest: 10:20 PM Today



Automations Logs

In the [Blynk.Console](#), you may also view the Automation Logs while editing an Automation. This option is not available in the [Blynk.App](#).

The image shows two screenshots from the Blynk Console. The left screenshot is the 'New Automation' screen, where a red circle highlights the 'Automation Logs' icon (a document with a checkmark) in the top right corner. The right screenshot shows the 'Automation Logs' page, which displays a table of automation execution history.

Automation Logs

1h 6h 1d 1w 1mo 3mo

Time	Device	Datastream	Value	Type
10:20:22 PM Today				Push

Selecting the 'Automation Logs' icon will display a page with a history of the executions of the Automation. The summary includes the date/time when it was executed, from what [Device](#), the Datastream employed, and the value of the [Datastream](#).

